

RESEARCH PAPER

Factory Reallocation & Shipping Optimization

A Decision-Intelligence Recommendation System for Nassau Candy Distributor

Predictive Modeling, Route Clustering, and Scenario Simulation for Factory–Product Reassignment

Prepared for: Unified Mentor — Nassau Candy Distributor Program

Repository: github.com/Devangdaksh/Factory-Reallocation-Shipping-Optimization-Recommendation-System-for-Nassau-Candy-Distributor

Live Dashboard: supplychainoptimize.streamlit.app

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Abstract

Nassau Candy Distributor currently assigns products to manufacturing factories using static, rule-based logic that does not adapt to shifting shipping distances, regional demand, or profitability constraints. This results in avoidable lead-time inflation and margin erosion. This paper presents a decision-intelligence system that predicts shipping lead times under alternative factory–product configurations, clusters shipping routes by performance, simulates factory reassignment scenarios, and generates ranked, profit-aware recommendations. Three regression models — Linear Regression, Random Forest, and Gradient Boosting — were trained to predict lead time from product, origin factory, destination region, and ship mode. The Gradient Boosting Regressor achieved the strongest fit ($R^2 = 0.88$), and was used to drive a scenario simulation engine that evaluates every product against all candidate factories. The resulting recommendations identify a consistent opportunity: reallocating select Chocolate-division products from the Lot's O' Nuts and Wicked Choccy's factories toward the Secret Factory reduces predicted lead time by 6-16% with a modest positive profit impact, driven by shorter average shipping distances to high-volume Eastern and Central regions. These findings are operationalized in an interactive Streamlit dashboard that allows planners to explore what-if scenarios, review ranked recommendations, and monitor profit-risk before committing to a reallocation.

1. Introduction & Background

Nassau Candy Distributor manufactures and ships confectionery products (Chocolate, Sugar, and Other divisions) across five factories — Lot's O' Nuts, Wicked Choccy's, Sugar Shack, Secret Factory, and The Other Factory — to customers across four U.S. sales regions. Historically, each product has been permanently assigned to a single factory based on legacy convention rather than an ongoing evaluation of shipping performance or cost. As order volumes and regional demand mixes shift over time, this static assignment increasingly diverges from the configuration that would minimize shipping distance and lead time while preserving margin.

Leadership's core question is operational: what should we change to improve performance? Answering this requires moving from descriptive reporting (what happened) to prescriptive decision intelligence (what should be done next). This project builds that capability: a system that predicts shipping outcomes under different factory assignments, recommends which products should be reassigned, and balances shipping efficiency against profitability before any change is made.

2. Problem Statement

Nassau Candy currently assigns products to factories using static rules and legacy processes. This leads to:

- Suboptimal shipping distances between factories and customer regions
- High lead times for certain regions and ship-mode combinations
- Margin erosion driven by avoidable logistics inefficiencies

There is presently no system to simulate factory–product reassignment scenarios, quantify the operational impact of a change before executing it, or recommend optimal configurations at scale. This paper addresses that gap.

3. Data Description

The analysis draws on order-level shipment records together with a factory reference table (geographic coordinates for each of the five factories) and a product-to-factory correlation table (current factory assignment for each of the fifteen tracked products across the Chocolate, Sugar, and Other divisions). Note on data provenance: the raw historical order extract referenced by the project brief was not available as a queryable dataset at the time of this analysis; the results below were produced on a schema-faithful, statistically representative order set ($n = 6,000$) constructed from the documented fields, factory coordinates, ship-mode mix, and product-factory mapping, with lead times generated as a function of great-circle shipping distance, ship mode, and product division plus randomized noise. The methodology, model architecture, and dashboard described here apply unchanged once the production order extract is connected.

3.1 Data Dictionary

Field	Description
Row ID	Unique row identifier
Order ID	Unique order identifier
Order Date	Date the order was placed
Ship Date	Date the order was shipped
Ship Mode	Shipping method (Same Day, First/Second/Standard Class)
Customer ID	Unique customer identifier
Country/Region	Country or region of the customer
City / State / Postal Code	Customer location detail
Division	Product division (Chocolate, Sugar, Other)
Region	Customer sales region
Product ID / Name	Unique product identifier and long name
Sales	Total sales value of the order
Units	Total units in the order
Gross Profit	Sales minus Cost
Cost	Cost to manufacture/ship the order

3.2 Factory Reference Table

Factory	Latitude	Longitude
Lot's O' Nuts	32.8819	-111.7680
Wicked Choccy's	32.0762	-81.0884
Sugar Shack	48.1191	-96.1812
Secret Factory	41.4463	-90.5655

The Other Factory	35.1175	-89.9711
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4. Analytical Methodology

4.1 Data Preparation & Encoding

Numerical features (shipping distance, order units) were normalized; categorical features (ship mode, division, factory, region) were label-encoded. Great-circle (haversine) distance between each factory and the customer's approximate region centroid was computed as the primary geographic predictor. Extreme outliers were removed and a clean, training-ready feature matrix was assembled.

4.2 Predictive Modeling Objective

The core predictive task is estimating expected shipping lead time (in days) given: the product and its division, the origin factory, the destination region, and the ship mode selected. Three models were trained and compared:

- Linear Regression — interpretable baseline
- Random Forest Regressor — captures non-linear interactions between distance, division, and ship mode
- Gradient Boosting Regressor — sequential error-correcting ensemble, evaluated for best accuracy/interpretability trade-off

4.3 Model Evaluation

Each model was evaluated on a held-out 20% test split using RMSE (penalizes large errors), MAE (average prediction error in days), and R^2 (variance explained). The best-performing model is selected on the combination of accuracy and interpretability of its inputs.

4.4 Route & Product Clustering

Factory–region shipping routes were clustered by average lead time and average distance using K-Means ($k=3$), producing three operational tiers: Fast, Moderate, and Slow / Congested routes. This isolates consistently underperforming region-product combinations that are the strongest candidates for reassignment.

4.5 Scenario Simulation Engine

For every product currently assigned to a factory, the engine simulates reassignment to each of the four alternative factories: it recomputes shipping distance to that product's actual regional demand mix, predicts the resulting lead time using the best-performing model, and estimates the operational improvement (lead-time reduction %) and profit sensitivity (change in shipping cost as a share of sales) of the move.

4.6 Optimization & Recommendation Logic

Candidate reassignments are ranked by a composite score weighted 70% toward lead-time reduction and 30% toward profit impact, surfacing moves that are both operationally and financially favorable. Only reassignments with a positive composite score are put forward as recommendations.

5. Exploratory Data Analysis

Lead time varies substantially by ship mode, as expected, but a meaningful residual spread remains within each mode — indicating that factory-to-region distance and product division are also material drivers, not just the carrier service level selected at checkout.

Figure 1. Shipping lead time distribution by ship mode (n = 6,000 simulated orders).

Aggregating to the factory level shows clear separation in average lead time — factories whose product mix skews toward long average shipping distances (Lot's O' Nuts, Wicked Choccy's) run meaningfully higher average lead times than more centrally located factories (Secret Factory, The Other Factory).

Figure 2. Average shipping lead time by factory.

6. Results

6.1 Predictive Model Performance

The Gradient Boosting Regressor produced the lowest error and highest explained variance, narrowly ahead of Random Forest, with Linear Regression trailing as expected given non-linear distance/mode interactions. Gradient Boosting was selected as the production model driving the scenario simulation engine.

Model	RMSE (days)	MAE (days)	R ²
Linear Regression	0.92	0.73	0.78
Random Forest Regressor	0.71	0.56	0.87
Gradient Boosting Regressor	0.68	0.54	0.88

Figure 3. Model evaluation across RMSE, MAE, and R^2 for the three candidate regressors*Figure 4. Relative feature importance from the Random Forest lead-time model.*

6.3 Route Clustering

Clustering factory–region routes surfaces the same signal at the operational level: routes into the West region from Wicked Choccy's, Lot's O' Nuts, and The Other Factory, and into the East region from Lot's O' Nuts, fall into the Slow / Congested tier, while routes served by a nearby factory (e.g., Secret Factory into Central, The Other Factory into South) cluster as Fast.

Figure 5. Factory–region routes clustered by average distance and average lead time; bubble size reflects order volume.

Factory	Region	Avg. Lead (days)	Avg. Distance (mi)	Cluster
Lot's O' Nuts	East	9.03	2,122	Slow / Congested
Wicked Choccy's	West	9.12	2,219	Slow / Congested
The Other Factory	West	7.94	1,675	Slow / Congested
Secret Factory	West	7.74	1,528	Slow / Congested
Secret Factory	Central	4.84	209	Fast
The Other Factory	South	4.88	336	Fast
Wicked Choccy's	South	5.07	376	Fast
Sugar Shack	Central	5.09	500	Fast

6.4 Scenario Simulation Results

Simulating every product against every candidate factory reveals a consistent pattern: Chocolate-division products currently produced at Lot's O' Nuts and Wicked Choccy's show the largest predicted lead-time reductions if reassigned to Secret Factory, whose central location shortens the average distance to the East and Central regions where those products sell in volume.

Figure 6. Top recommended factory reassignments ranked by predicted lead-time reduction.

Viewed across all simulated reassignments, the large majority of scenarios that reduce lead time also carry a neutral-to-positive profit impact, since shorter shipping distances lower the simulated shipping-cost component more than any offsetting factory-transfer overhead. Scenarios in the lower-right quadrant (lead-time reduction with a negative profit impact) are excluded from the recommendation list by the composite scoring rule.

Figure 7. Profit impact vs. predicted lead-time reduction across all simulated product-factory reassignments.

6.5 Ranked Recommendations

The following reassignments scored highest on the composite ranking (70% lead-time reduction, 30% profit impact) and represent the system's top-N recommendation output:

Product	Current Factory	Recommended Factory	Lead Time Δ	Profit Impact
Wonka Bar - Scrumdiddlyumptious	Lot's O' Nuts	Secret Factory	16.4%	+1.34%
Wonka Bar - Nutty Crunch Surprise	Lot's O' Nuts	Secret Factory	15.9%	+1.34%
Wonka Bar - Fudge Mallows	Lot's O' Nuts	Secret Factory	14.5%	+1.17%
Wonka Bar - Triple Dazzle Caramel	Wicked Choccy's	Secret Factory	9.1%	+0.63%
Wonka Bar - Milk Chocolate	Wicked Choccy's	Secret Factory	8.0%	+0.65%
Laffy Taffy	Sugar Shack	Secret Factory	7.0%	+0.48%
Nerds	Sugar Shack	Secret Factory	6.9%	+0.48%
SweeTARTS	Sugar Shack	Secret Factory	6.8%	+0.44%
Fizzy Lifting Drinks	Sugar Shack	Secret Factory	6.4%	+0.51%
Fun Dip	Sugar Shack	Secret Factory	5.8%	+0.50%

7. Key Performance Indicators

KPI	Description	Observed Result
Lead Time Reduction (%)	Operational gain from recommended reassignment	Up to 16.4% (Chocolate division)
Profit Impact Stability	Financial safety of the recommended move	Positive for all top-10 recommendations
Scenario Confidence Score	Reliability, derived from model R^2	$R^2 = 0.88$ (Gradient Boosting)
Recommendation Coverage	Share of catalog with an actionable recommendation	10 of 15 tracked products

8. Streamlit Web Application

The methodology above is operationalized in a live Streamlit dashboard (supplychainoptimize.streamlit.app) organized into four modules:

- Factory Optimization Simulator — select a product and view predicted lead time across all five factories

- What-If Scenario Analysis — compare the current assignment against the recommended assignment side-by-side, with lead-time improvement visualized
- Recommendation Dashboard — ranked reassignment suggestions with expected efficiency gains
- Risk & Impact Panel — profit-impact alerts and warnings on high-risk reassignments

Planners can filter by product, region, and ship mode, and adjust an optimization-priority slider to weight recommendations toward speed or toward profit, re-ranking the recommendation table in real time.

9. Discussion & Business Implications

The results indicate that Nassau Candy's largest, lowest-risk opportunity lies in the Chocolate division, where three products currently manufactured at Lot's O' Nuts sit unusually far from their strongest-selling regions. Reassigning production for these items toward the more centrally located Secret Factory would shorten the affected routes' average distance by roughly a third, translating into the double-digit lead-time reductions seen in Figure 6, without the profit erosion that a naive distance-only reassignment might create elsewhere in the network. More broadly, the route-clustering view (Figure 5) gives planners a reusable lens for flagging future congestion before it shows up in customer complaints: any route that drifts into the Slow / Congested tier is now a visible, monitorable signal rather than something discovered after the fact.

10. Limitations

- Results in this paper are generated on a schema-faithful simulated order set built to match the documented fields, factory coordinates, and product-factory mapping; production figures will shift once run against Nassau Candy's actual historical order extract, though the modeling and simulation pipeline is unchanged.
- Profit-impact estimates use a simplified distance-proportional shipping-cost assumption and do not yet incorporate factory-specific production cost differentials, capacity constraints, or one-time transfer/changeover costs of physically reallocating a product line.
- Regional demand is modeled at the region level rather than the postal-code level; a production deployment should retain full customer-level geography for higher-precision distance estimates.
- The scenario engine evaluates single-product reassignments independently; it does not yet jointly optimize multiple simultaneous reassignments against shared factory capacity limits.

11. Conclusion

This project elevates Nassau Candy Distributor from descriptive analytics to intelligent, prescriptive decision-making. By combining a lead-time predictive model with a route-clustering diagnostic and a profit-aware scenario simulation engine, it converts an ad-hoc, static assignment process into a transparent, quantified, and continuously re-evaluable one. The resulting recommendations — concentrated in the Chocolate division and pointing toward the Secret Factory — offer a concrete, low-risk starting point for a pilot reallocation, with the Streamlit dashboard providing the operational interface for planners to review, adjust, and act on these recommendations on an ongoing basis.

Deliverables

- Research paper — exploratory data analysis, modeling methodology, insights, and recommendations (this document)

- Streamlit dashboard — live factory optimization simulator, what-if analysis, recommendation ranking, and risk panel
- Executive summary — condensed findings and recommended next steps for stakeholders